

Week 4 – Statistics, Data Visualization & EDA

Internova Data Analysis Internship

Student Performance Dataset

Task 1 – Mean, Median & Mode

Practical output from the student performance analysis.

TASK 1 — MEAN, MEDIAN & MODE

Marks: [np.int64(65), np.int64(70), np.int64(75), np.int64(68), np.int64(82), np.int64(74), np.int64(90), np.int64(60), np.int64(80), np.int64(94)]

Mean: 75.80

Median: 74.50

Mode: No unique mode

Task 2 – Variance & Standard Deviation

Practical output from the student performance analysis.

TASK 2 — VARIANCE & STANDARD DEVIATION

Variance: 105.36

Standard Deviation: 10.26

Task 3 – Correlation & Probability Basics

Practical output from the student performance analysis.

TASK 3 — CORRELATION & PROBABILITY

Correlation (Study Hours vs Marks): 0.99

Relationship: Positive

Probability of a student scoring 75 or more: $5/10 = 0.50 = 50\%$

Task 4 – Outlier Detection

Practical output from the student performance analysis.

TASK 4 — OUTLIER DETECTION (IQR)

Q1: 68.50

Q3: 84.25

IQR: 15.75

Lower limit: 44.88

Upper limit: 107.88

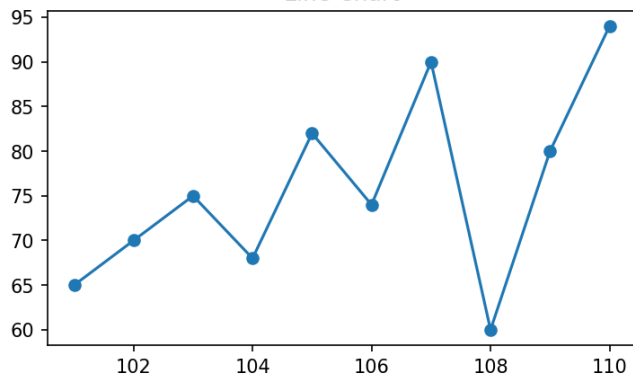
Outlier(s): [np.int64(150)]

Task 5 – Matplotlib Visualization

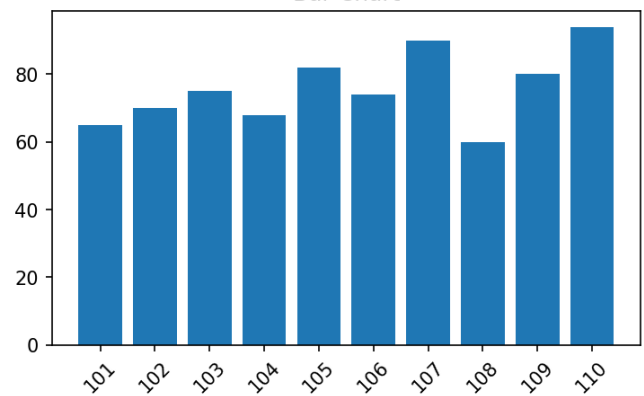
Practical output from the student performance analysis.

TASK 5 — Matplotlib Visualizations

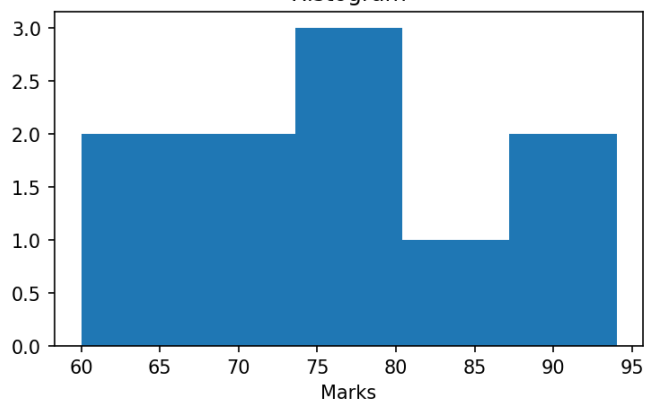
Line Chart



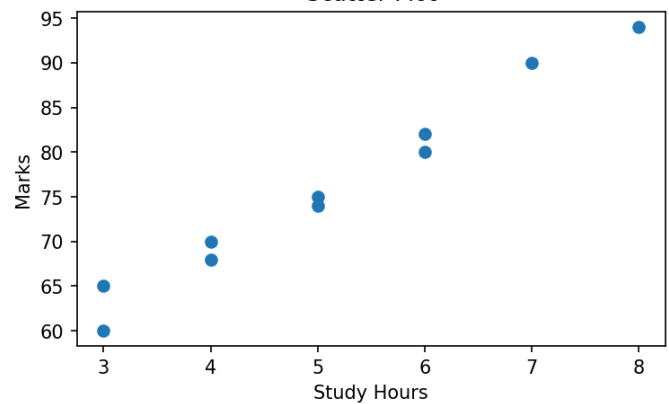
Bar Chart



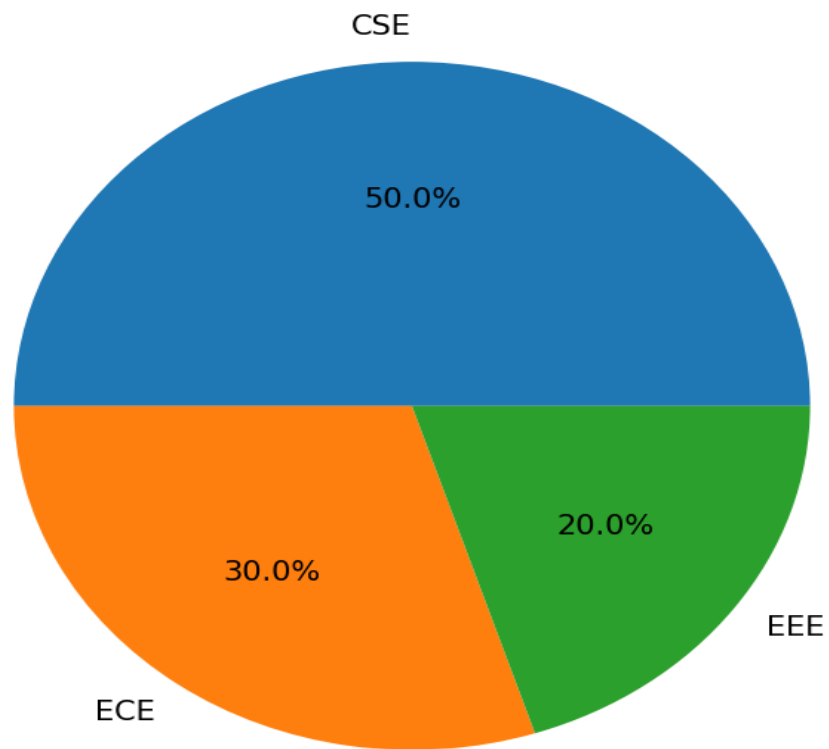
Histogram



Scatter Plot



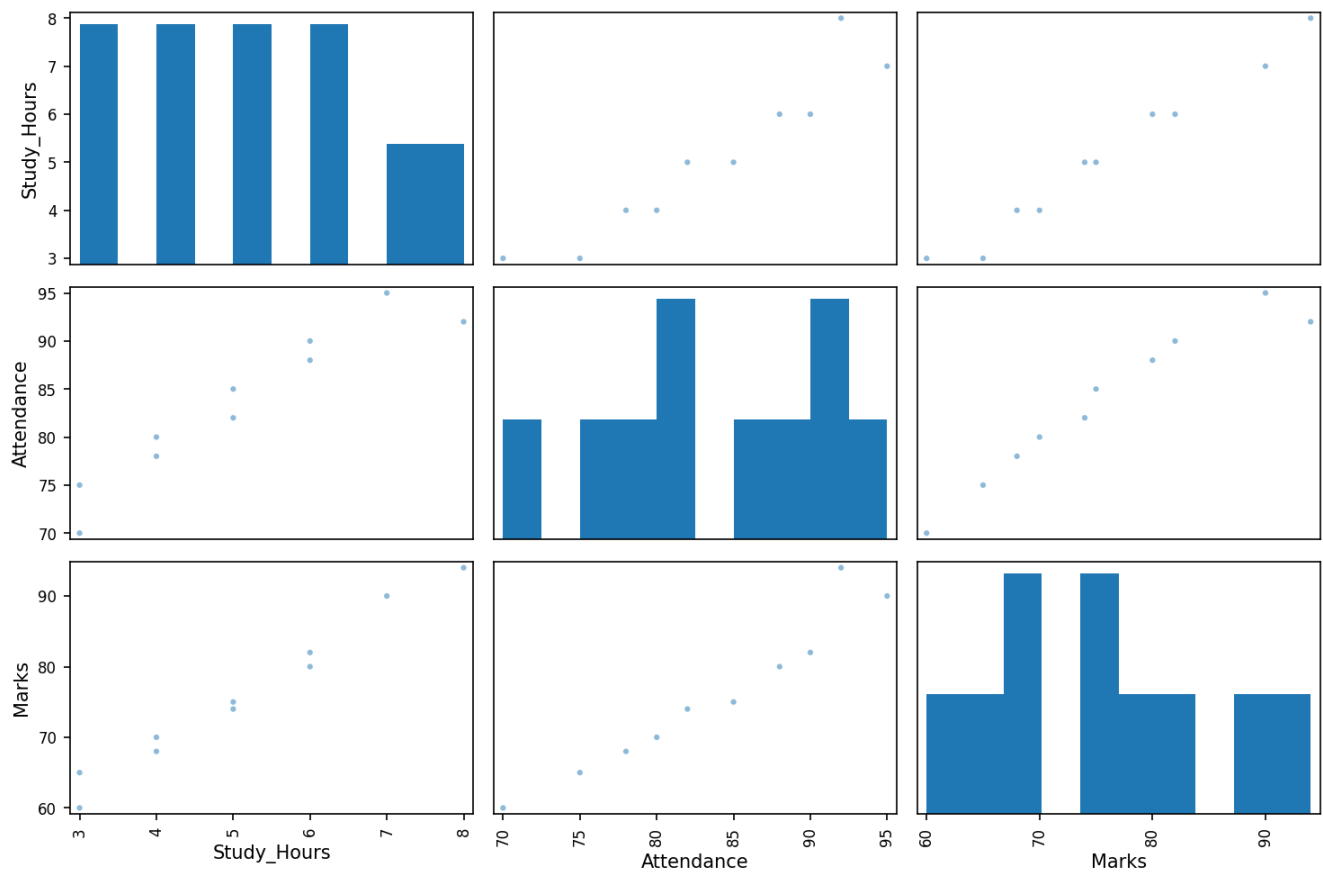
Pie Chart — Students by Branch



Task 6 – Seaborn Visualization

Practical output from the student performance analysis.

TASK 6 — Seaborn / Relationship Visualizations



Task 7 – EDA Data Inspection & Cleaning

Practical output from the student performance analysis.

TASK 7 — BEFORE CLEANING

Student_ID	Name	Branch	Study_Hours	Attendance	Marks
101	Arun	CSE	3.0	75.0	65.0
102	Rahul	ECE	4.0	80.0	70.0
103	Priya	CSE	NaN	85.0	75.0
104	Sneha	EEE	4.0	78.0	68.0
105	Kiran	CSE	6.0	NaN	82.0
106	Ravi	ECE	5.0	82.0	74.0
107	Anjali	CSE	7.0	95.0	NaN
108	Vijay	EEE	3.0	70.0	60.0
109	Neha	ECE	6.0	88.0	80.0
110	Manoj	CSE	8.0	92.0	94.0
107	Anjali	CSE	7.0	95.0	NaN

Missing values:

Student_ID 0
Name 0
Branch 0
Study_Hours 1
Attendance 1
Marks 2

Duplicate rows: 1

TASK 7 — AFTER CLEANING

Student_ID	Name	Branch	Study_Hours	Attendance	Marks
101	Arun	CSE	3.000000	75.000000	65.000000
102	Rahul	ECE	4.000000	80.000000	70.000000
103	Priya	CSE	5.111111	85.000000	75.000000
104	Sneha	EEE	4.000000	78.000000	68.000000
105	Kiran	CSE	6.000000	82.777778	82.000000
106	Ravi	ECE	5.000000	82.000000	74.000000
107	Anjali	CSE	7.000000	95.000000	74.222222
108	Vijay	EEE	3.000000	70.000000	60.000000
109	Neha	ECE	6.000000	88.000000	80.000000
110	Manoj	CSE	8.000000	92.000000	94.000000

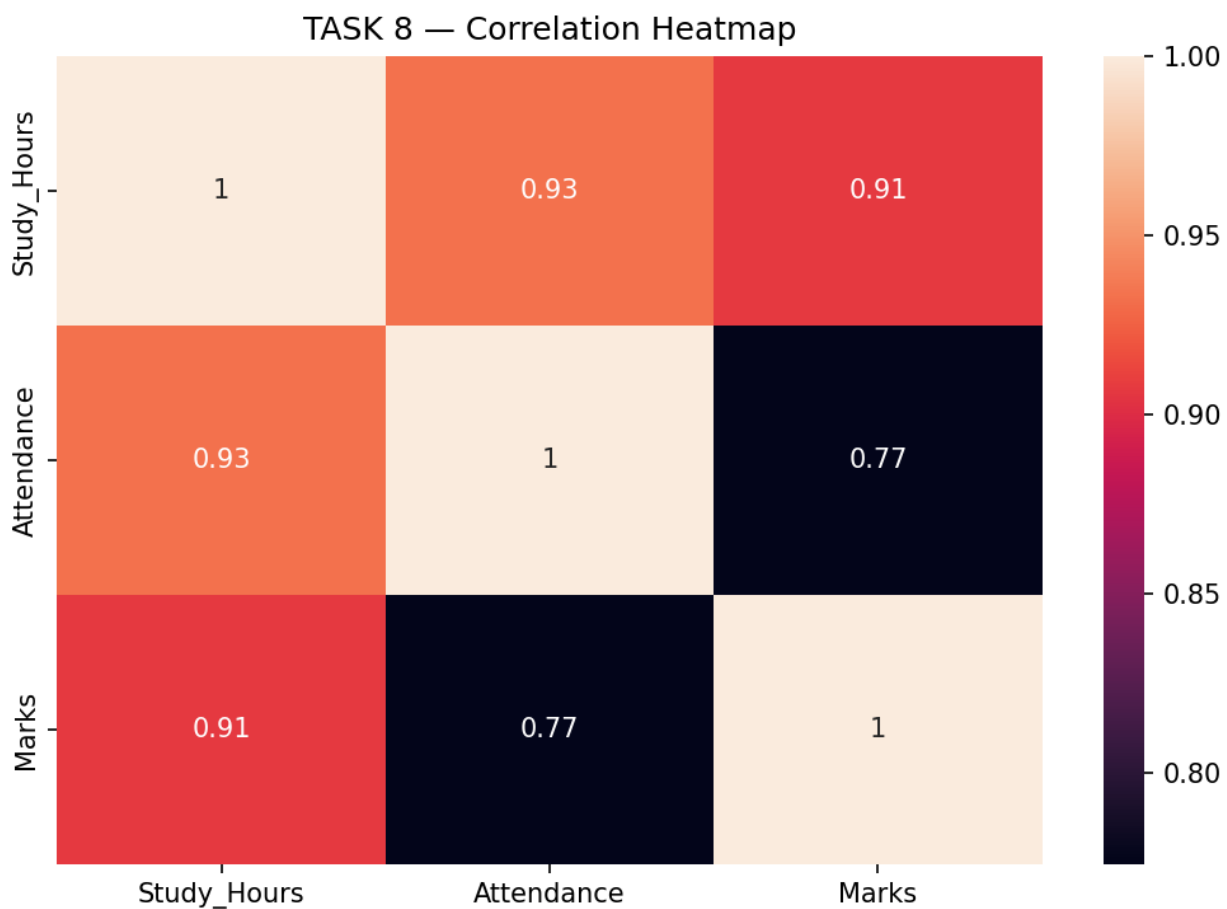
Missing values after cleaning:

Student_ID 0
Name 0
Branch 0
Study_Hours 0
Attendance 0
Marks 0

Duplicate rows after cleaning: 0

Task 8 – EDA Correlation & Insights

Practical output from the student performance analysis.



TASK 8 — EDA INSIGHTS

Average marks by branch:

Branch

CSE 78.04

ECE 74.67

EEE 64.00

Average study hours: 5.11

Average attendance: 82.78

Average marks: 74.22

1. Study hours show a positive relationship with marks.
2. Attendance shows a positive relationship with marks.
3. Average marks differ across branches.

Task 9 – Business Recommendations

Practical output from the student performance analysis.

TASK 9 — BUSINESS RECOMMENDATIONS

1. Encourage consistent study time based on the positive relationship between study hours and marks.
2. Monitor student attendance based on the positive relationship between attendance and marks.

These recommendations are specific to this small practice dataset.

Conclusion

This project covers basic statistics, visualization, data cleaning, exploratory analysis, insights and business recommendations. Findings are specific to the selected practice dataset.